

Machine Learning Workflow and Process

Successful machine learning projects follow structured workflows balancing art and science. Problem definition establishes clear objectives, identifying whether the problem suits ML solutions and defining success metrics. Data collection gathers relevant examples capturing phenomena of interest. Data quality and quantity significantly impact results—insufficient or poor-quality data limits model performance. Exploratory data analysis examines data distributions, identifies patterns, reveals anomalies, and informs feature engineering. Feature engineering transforms raw data into meaningful representations the model can learn from effectively. Domain expertise guides feature creation. Data preprocessing addresses missing values, outliers, and normalizes feature scales ensuring algorithms function optimally. Data splitting partitions data into training sets for learning, validation sets for tuning hyperparameters, and test sets for unbiased performance evaluation. Model selection chooses appropriate algorithms for the problem. Training fits the model to training data, minimizing loss functions through optimization. Validation assesses model performance on unseen data, detecting overfitting where training performance exceeds validation performance. Hyperparameter tuning optimizes parameters controlling learning behavior. Evaluation on test data provides final performance estimates. Model interpretation reveals learned patterns and decision factors. Deployment integrates models into production systems. Monitoring tracks performance over time, detecting data drift where input distributions change.